

Chapter 10: Independence and Conditional Probabilities

The Practice of Statistics in the Life Sciences
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Part II: Probability and Distributions

Introduction

In Chapter 9, we often studied one random phenomenon at a time. In practice, biological and scientific events are usually connected.

- Does a particular gene alter the probability of developing a disease?
- If a diagnostic test is positive, what is the probability the disease is actually present?
- Does knowing one event occurred change the probability of another?

Chapter Goal

Extend our probability toolkit from single events to **intersections, dependence, conditioning, sequential events and reverse probabilities**.

Chapter Roadmap: Four Questions to Keep Separate

1. **Can two events happen together?**

Intersections, disjoint events, and Venn diagrams.

2. **Does knowing one event occurred change the probability of another?**

Independence and dependence.

3. **What is the probability after new information is given?**

Conditional probability.

4. **How do we reverse a conditional probability?**

Tree diagrams, the Law of Total Probability, and Bayes's Theorem.

Sample Spaces and Events

A **sample space** is the set of all possible outcomes. An **event** is a set of outcomes within that sample space.

Example: Roll a fair six-sided die

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let A be the event “the result is even” and let B be “the result is greater than 3”.

$$A = \{2, 4, 6\} \quad B = \{4, 5, 6\}$$

Because the six outcomes are equally likely, the probability of an event is

$$P(\text{event}) = \frac{\text{number of outcomes in the event}}{\text{number of outcomes in } S}$$

For example $P(A) = \frac{3}{6} = \frac{1}{2}$.

Combining Events as Sets

Keep $S = \{1, 2, 3, 4, 5, 6\}$, $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$

Operation	Outcomes	Probability
$A \cap B$: even and greater than 3	$\{4, 6\}$	$\frac{2}{6}$
$A \cup B$: even or greater than 3	$\{2, 4, 5, 6\}$	$\frac{4}{6}$
A^c : not even	$\{1, 3, 5\}$	$\frac{3}{6}$

How to read “or”

The union $A \cup B$ includes outcomes in A , in B or in both. We count 4 and 6 only once.

Since $A \cap B = \{4, 6\}$, the events overlap: they are not disjoint.

Probability Notation

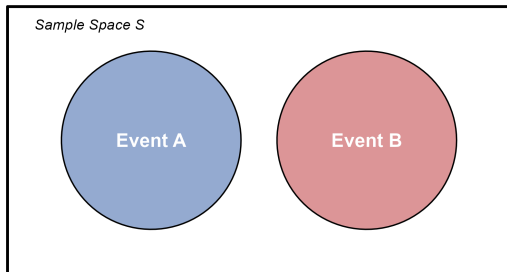
Language	Notation	Meaning
A and B	$A \cap B$	Both events occur
A or B	$A \cup B$	At least one occurs; “or” is inclusive
not A	A^c	Event A does not occur
B given A	$P(B \mid A)$	Probability of B after restricting attention to cases where A occurred

Before Calculating

Translate the question into **and**, **or**, **not** or **given**.

Visualizing Multiple Events: Disjoint Events

Disjoint Events (Mutually Exclusive)



Disjoint Events

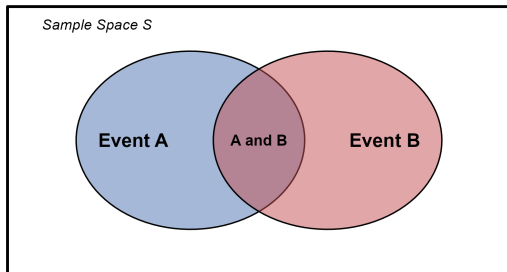
Two events are **disjoint** if they cannot occur together.

$$P(A \cap B) = 0$$

The circles do not overlap, so there is no outcome belonging to both events.

Visualizing Multiple Events: Overlapping Events

Events That Are NOT Disjoint



Most interesting scientific events are **not disjoint**.

- A patient can have high blood pressure and high cholesterol.
- The overlap represents $A \cap B$.
- A nonempty overlap can have positive probability.

Example 10.1: Two Births

Assume

- Each birth has probability 0.5 of being a girl;
- The two birth outcomes are modelled as independent.

Question

Under the simplified model, what is the probability that the next two births are both girls?

Define

A = first baby is a girl B = second baby is a girl

Translate the Question

“Both births are girls” means the intersection $A \cap B$.

Example 10.1: Two Births

	Second is B	Second is B^c
First is A	$A \cap B$	$A \cap B^c$
First is A^c	$A^c \cap B$	$A^c \cap B^c$

Under the simplified independent model, the second birth still has probability 0.5 of being a girl after we know the first birth was a girl.

Why Multiply?

To reach the “girl then girl” cell, the first event must occur **and then** the second event must occur. Because the events are independent, we multiply their probabilities.

$$P(A \cap B) = P(A)P(B) = (0.5)(0.5) = \mathbf{0.25}$$

So the model assigns a **25% probability** to two girls in two births.

The Gambler's Fallacy and Independence

Independence

Events are independent when knowing the outcome of one does not change the probability of the other.

Gambler's Fallacy

The gambler's fallacy is the mistaken belief that an outcome becomes more likely because it has occurred less often than expected. For example, if a fair die has not shown a 4 in several rolls, someone might think a 4 is “due.” Each roll is independent, however, so the probability of rolling a 4 next remains $\frac{1}{6}$.

Long-run proportions may stabilize but future outcomes do not need to compensate for a past streak.

Independent or Dependent?

- Two coin tosses
- Two cards drawn without replacement
- Taking the same cognitive test twice
- Repeated random measurement errors

Define the Events Carefully

Independence applies to the **events or random quantities you define**. Similar measurements are not automatically dependent and separate measurements are not automatically independent.

Multiplication Rule for Independent Events

Two Events

If A and B are independent

$$P(A \cap B) = P(A)P(B)$$

Three Events

If A , B , and C are **mutually independent**

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

Important Caveat

Pairwise independence alone is **not sufficient** to guarantee the three event product rule.

Do Not Assume Independence by Default

The product rule $P(A \cap B) = P(A)P(B)$ is a shortcut with a condition attached. Use it when:

- independence is explicitly stated
- independence is justified by the data-generating process
- independence has been established mathematically

Example 10.2: Rapid HIV Testing

Question

A clinic tests 200 HIV-negative people. Suppose the false-positive probability for one HIV negative person is 0.004. What is the probability that **at least one** of the 200 tests produces a false positive? For this calculation, assume the test outcomes for different people are independent.

Look for the Complement

The opposite of “at least one false positive” is much simpler: **no false positives at all**.

Example 10.2: Rapid HIV Testing

For one HIV-negative person,

$$P(\text{correct negative}) = 1 - 0.004 = 0.996$$

Because the 200 test outcomes are assumed independent,

$$P(\text{all 200 correct negatives}) = (0.996)^{200} = 0.4486$$

Now use the complement rule:

$$P(\text{at least one false positive}) = 1 - 0.4486 = \mathbf{0.5514}$$

Interpretation

Even though a false positive is rare for any one HIV-negative person, repeating the test across many people creates many opportunities for at least one error. Under this model, the chance is about **55.1%**.

Example 10.2: Rapid HIV Testing



- For one HIV-negative person, the true-negative rate is 99.6%.
- Across 200 independent tests, the chance of at least one false positive is about 55.1%.

Disjoint vs. Independent

Disjoint

Cannot occur together:

$$P(A \cap B) = 0$$

If both have positive probability, observing A tells us that B did not occur.

Independent

Knowing one event occurred does not change the probability of the other:

$$P(A \cap B) = P(A)P(B)$$

Precise Statement

Two **positive probability** disjoint events cannot be independent. A zero probability event is the mathematical exception to the blanket statement “disjoint events are never independent.”

From Special Rules to General Rules

The familiar shortcuts have conditions:

- $P(A \cup B) = P(A) + P(B)$ only when A and B are disjoint.
- $P(A \cap B) = P(A)P(B)$ only when A and B are independent.

General Rules

We now remove those restrictions so the formulas work for **any two events**.

The General Addition Rule

If A and B overlap, adding $P(A)$ and $P(B)$ counts the intersection twice.

For Any Two Events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A and B are disjoint then $P(A \cap B) = 0$ and the formula reduces to

$$P(A \cup B) = P(A) + P(B)$$

Example 10.5: Dalmatians

Let

- I = the dog has hearing impairment
- B = the dog has blue eyes.

Suppose

$$P(I) = 0.28 \quad P(B) = 0.11 \quad P(I \cap B) = 0.05$$

Question

What is the probability that a randomly selected Dalmatian is hearing-impaired **or** blue-eyed?

The word “or” means the union $I \cup B$. If we simply add $0.28 + 0.11$, dogs in the overlap are counted once as impaired and a second time as blue-eyed.

Example 10.5: Dalmatians

Use the General Addition Rule

$$P(I \cup B) = P(I) + P(B) - P(I \cap B)$$

Substitute the data

$$P(I \cup B) = 0.28 + 0.11 - 0.05 = \mathbf{0.34}$$

So 34% are in at least one of the two categories. To find the probability of **neither**, use the complement:

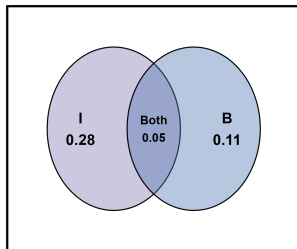
$$P(\text{neither}) = 1 - 0.34 = \mathbf{0.66}$$

Interpretation

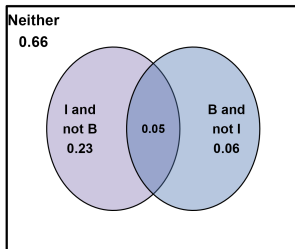
The subtraction does not remove the overlap from the final answer. It removes the **extra copy** created by adding the two marginal probabilities.

Example 10.5: Dalmatians – Visualizing the General Addition Rule

(a) Simple Probabilities of I and B



(b) Disjoint Combinations (Sum to 1)



The second diagram partitions the sample space into disjoint pieces:

$$0.23 + 0.05 + 0.06 + 0.66 = 1$$

Partition of an Event

Every dog with hearing impairment either has blue eyes or does not. These two groups cannot overlap.

Partition of an event

For any events A and B

$$A = (A \cap B) \cup (A \cap B^c)$$

Because the two parts are disjoint,

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

In the Dalmatian example, I is hearing impairment and B is blue eyes. The Venn diagram gives

$$P(I) = P(I \cap B) + P(I \cap B^c) = 0.05 + 0.23 = 0.28$$

The 0.23 represents dogs with hearing impairment **and without** blue eyes.

Conditional Probability

Conditional probability answers a question after we are given new information.

Formula

When $P(A) > 0$

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

- A is the **condition**: the information after the bar.
- B is the event whose probability we want.
- Conditioning on A makes A our **new denominator or restricted sample space**.

Example 10.3: Forest Data

Ecologists classify trees by species and life stage. The entries below are proportions of the **entire forest**.

	Dead	Live	Sapling	Total
Western red cedar (WRC)	0.02	0.10	0.08	0.20
Douglas fir (DF)	0.16	0.16	0.00	0.32
Western hemlock (WH)	0.23	0.21	0.04	0.48
Total	0.41	0.47	0.12	1.00

Question

A randomly selected tree is known to be a **sapling**. What is the probability that it is a western hemlock?

Example 10.3: Forest Dat

Before using the new information, a randomly selected tree has

$$P(WH) = 0.48$$

This 0.48 uses the **entire forest** as the denominator

What Changes?

The word “sapling” is the condition. Once we know the selected tree is a sapling, the whole forest is no longer the relevant denominator

We now restrict attention to the sapling column, which contains 12% of the forest. The question becomes a **within sapling proportion**

Example 10.3: Forest Data

The condition “sapling” restricts our attention to the sapling column.

- Saplings make up 0.12 of the entire forest.
- Western-hemlock saplings make up 0.04 of the entire forest.

Therefore, among the saplings

$$P(WH \mid \text{sapling}) = \frac{P(WH \cap \text{sapling})}{P(\text{sapling})} = \frac{0.04}{0.12} = \mathbf{0.3333}$$

Interpretation

About **33.3% of saplings** are western hemlocks. This is different from saying that 48% of all trees are western hemlocks.

Denominator Check

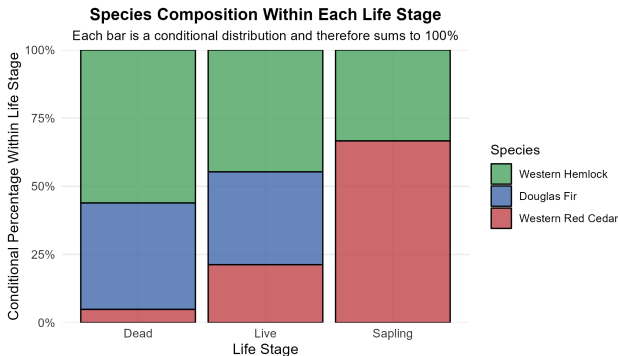
In $P(B \mid A)$, the condition A tells you which group becomes the new denominator.

Example 10.3: Forest Data

Let D , L and S mean dead, live and sapling. Let WRC, DF and WH denote the three species.

	Given dead (D)	Given live (L)	Given sapling (S)
Western red cedar	$P(\text{WRC} \mid D) = 0.049$	$P(\text{WRC} \mid L) = 0.213$	$P(\text{WRC} \mid S) = 0.667$
Douglas fir	$P(\text{DF} \mid D) = 0.390$	$P(\text{DF} \mid L) = 0.340$	$P(\text{DF} \mid S) = 0$
Western hemlock	$P(\text{WH} \mid D) = 0.561$	$P(\text{WH} \mid L) = 0.447$	$P(\text{WH} \mid S) = 0.333$
Column total	1.000	1.000	1.000

Example 10.3: Forest Data



Conditioning changes the group we are looking at.

- Entire forest: western hemlock = 48%.
- Sapling column only: western hemlock = 33%.

Key Idea

Conditional probability is a **within group proportion**.

Example 10.4: Reversing the Condition

Question

Among **western hemlock trees**, what proportion are saplings?

Previously we asked the reverse conditional probability:

$$P(WH \mid \text{sapling}) = 0.3333$$

which means: *among saplings, what proportion are western hemlocks?*

The condition is now WH , so the western hemlock row becomes the denominator:

$$P(\text{sapling} \mid WH) = \frac{0.04}{0.48} = \mathbf{0.0833}$$

Why the Answers Differ

The numerator is the same intersection, 0.04, but the denominators are different groups: 0.12 versus 0.48.

Example 10.4: Reversing the Condition

$$P(WH \mid \text{sapling}) = 0.3333$$

$$P(\text{sapling} \mid WH) = 0.0833$$

$$P(WH \cap \text{sapling}) = 0.04$$

- $P(WH \mid \text{sapling})$: among **saplings**, the percentage that are WH.
- $P(\text{sapling} \mid WH)$: among **WH trees**, the percentage that are saplings.
- $P(WH \cap \text{sapling})$: among the **entire forest**, the percentage that are both.

Conditional Probability Habit

Read the vertical bar as “given”. Say the probability statement aloud before doing the calculation.

Reversing Conditional Probabilities

A small value of

$$P(\text{evidence} \mid \text{natural causes})$$

is not necessarily a small value of

$$P(\text{natural causes} \mid \text{evidence})$$

The prosecutor's fallacy

The prosecutor's fallacy occurs when someone treats the probability of seeing the evidence *if one explanation is true* as the probability that the explanation is true *given the evidence*. For example, “This evidence would be rare under natural causes” does not establish that natural causes are unlikely.

To assess the explanation after seeing the evidence, we also need to consider how likely the other explanations were and how often they produce the same evidence. Bayes's Theorem gives us a way to do this.

The General Multiplication Rule

Rearranging the conditional probability formula gives a rule valid whether or not events are independent.

For Any Two Events

$$P(A \cap B) = P(A)P(B | A)$$

Equivalently

$$P(A \cap B) = P(B)P(A | B)$$

Example 10.6: Soda Consumption and Weight

In a survey, suppose

$$P(\text{no soda}) = 0.52$$

and

$$P(\text{overweight} \mid \text{no soda}) = 0.39$$

The 39% is not 39% of the whole population. It is 39% **within the 52% subgroup** who report no soda.

Question

What proportion of **all adults** are both non-soda drinkers and self-described as overweight?

Therefore, we need the probability of travelling through both stages: first “no soda”, then “overweight within that group”.

Example 10.6: Soda Consumption and Weight — Multiply the Stages

Apply the General Multiplication Rule:

$$P(A \cap B) = P(A)P(B | A)$$

Here

$$P(\text{no soda} \cap \text{overweight}) = (0.52)(0.39) = \mathbf{0.2028}$$

Interpretation

Approximately **20.3% of all surveyed adults** are in both groups under these survey proportions.

Association Is Not Causation

This calculation describes a joint proportion. It does not show that avoiding soda causes weight status or that weight status causes soda consumption.

How to Mathematically Test Independence

Product Definition

Events A and B are independent exactly when

$$P(A \cap B) = P(A)P(B)$$

When $P(A) > 0$, an equivalent test is

$$P(B \mid A) = P(B)$$

Similarly, when $P(B) > 0$

$$P(A \mid B) = P(A)$$

Interpretation

Independence means that conditioning does **not change** the probability.

Example 10.9: Dalmatians

Question

Are hearing impairment and blue eyes independent in the Dalmatian data?

From the earlier Dalmatian data,

$$P(I) = 0.28, \quad P(B) = 0.11$$

Independence Test

If hearing impairment and blue eyes were independent, the expected joint probability would be

$$P(I)P(B) = (0.28)(0.11) = \mathbf{0.0308}$$

This 0.0308 is the overlap we would expect **if knowing one trait gave us no information about the other.**

Applications of Conditional Probability

The same calculation appears in many fields:

- **Chemistry:** probability an alloy passes a stress test given that it contains titanium.
- **Insurance:** probability a claim came from a younger driver, and claim rates within age groups.
- **Psychology:** probability of one clinical condition given a reported symptom.

Common Structure

Conditioning means we restrict attention to a specified subgroup and compute a proportion **within that group**.

Chemistry Example

A materials laboratory tests 500 alloys.

	Pass	Fail	Total
Contains titanium	120	30	150
No titanium	70	280	350
Total	190	310	500

Question

If an alloy is known to contain titanium, what is the probability that it passes the stress test?

Chemistry Example

The condition is “contains titanium” so we use only the **150 titanium alloys** as the denominator. Among the 150 titanium alloys, 120 pass:

$$P(\text{Pass} \mid Ti) = \frac{120}{150} = \mathbf{0.80}$$

For comparison, the overall pass rate is

$$P(\text{Pass}) = \frac{190}{500} = \mathbf{0.38}$$

Interpretation

The pass rate within the titanium group is 80%, compared with 38% overall. This is strong evidence of an association between titanium status and stress-test performance in these data.

Insurance Example

Suppose

$$P(U) = 0.20 \quad P(U \cap C) = 0.08 \quad P(C) = 0.16$$

where U means “under 25” and C means “filed a claim”. These statements mean:

- 20% of customers are under 25;
- 8% of all customers are both under 25 and claimants;
- 16% of all customers filed a claim.

Questions

1. Among customers who filed a claim, what proportion are under 25?
2. Among under-25 customers, what proportion filed a claim?

Insurance Example

The condition is “filed a claim”, so claimants form the new denominator:

$$P(U | C) = \frac{P(U \cap C)}{P(C)} = \frac{0.08}{0.16} = \mathbf{0.50}$$

Interpretation

Although under-25 customers are 20% of all customers, they make up 50% of the claimants in this hypothetical portfolio.

Insurance Example

Among under 25 customers

$$P(C | U) = \frac{0.08}{0.20} = \mathbf{0.40}$$

For customers age 25 or older,

$$P(C | U^c) = \frac{0.16 - 0.08}{1 - 0.20} = \frac{0.08}{0.80} = \mathbf{0.10}$$

Interpretation

In this hypothetical dataset, the observed claim rate is 40% for under 25 customers and 10% for customers 25 or older.

Actual insurance pricing uses many variables, actuarial models and regulatory requirements; age alone does not determine a premium.

Psychology Example: Question and Given Information

Suppose a survey reports

$$P(\text{Insomnia}) = 0.25$$

and

$$P(\text{Depression} \cap \text{Insomnia}) = 0.15$$

Question

Among people who report chronic insomnia, what proportion also satisfy the study's depression criterion?

Psychology Example: Calculate and Interpret Carefully

$$P(\text{Depression} \mid \text{Insomnia}) = \frac{P(\text{Depression} \cap \text{Insomnia})}{P(\text{Insomnia})} = \frac{0.15}{0.25} = \mathbf{0.60}$$

Interpretation

In this hypothetical survey population, 60% of people in the insomnia group also meet the stated depression criterion.

What This Does Not Mean

It is not an individual diagnosis and it does not show that insomnia causes depression or that depression causes insomnia.

Will the Beer Be Cold Before the Pizza Arrives?

At a physics department pizza night, the beer is about 18°C and the pizza will arrive in 20 minutes. Students test three ways to cool identical cans. They call a can **cold enough** if its temperature is at most 10°C after 20 minutes. Let C mean “cold enough.”

Cooling method	Proportion of trials	Chance of success
Refrigerator (R)	$P(R) = 0.30$	$P(C \mid R) = 0.10$
Freezer (F)	$P(F) = 0.30$	$P(C \mid F) = 0.55$
Ice-water bath (I)	$P(I) = 0.40$	$P(C \mid I) = 0.85$

Question

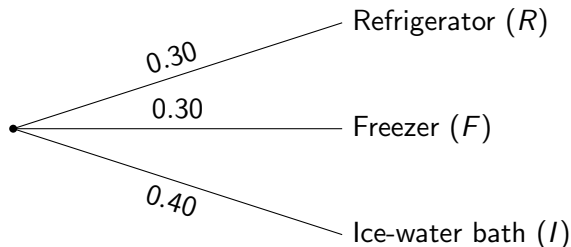
For a randomly selected trial, what is the probability that the beer is cold enough when the pizza arrives?

That is we want the $P(C)$. Recall partitioning of event; so we can write $P(C)$ as

$$P(C) = P(C \cap R) + P(C \cap F) + P(C \cap I)$$

Building the Tree: Which Cooling Method?

The first branches show which cooling method is used in a trial.



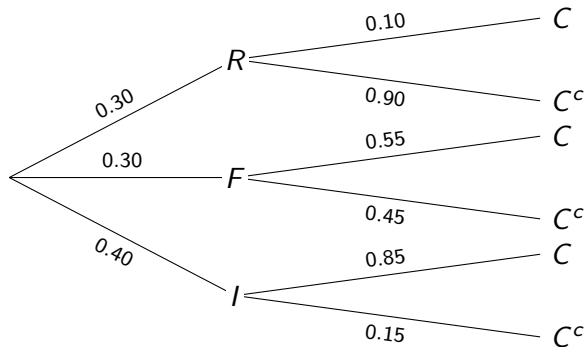
First tree rule

Each trial uses one of the three methods, so the branches leaving the starting point sum to 1.

$$P(R) + P(F) + P(I) = 0.30 + 0.30 + 0.40 = 1$$

Building the Tree: Is the Beer Cold Enough?

From each method, add branches for **cold enough** (C) and **not cold enough** (C^c).



The second branches are complement of conditional probabilities. For example, $P(C | I) = 0.85$, so $P(C^c | I) = 1 - P(C | I) = 0.15$.

Multiply Along a Complete Path

Consider the path **ice-water bath then cold enough**. The trial must use an ice-water bath *and* reach the target temperature.

General Multiplication Rule

$$P(I \cap C) = P(I)P(C | I) = (0.40)(0.85) = \boxed{0.34}$$

Thus, 34% of **all trials** use an ice-water bath and finish with beer at or below 10°C.

Second tree rule

Multiply along a complete path to find the probability of that combination of outcomes.

Add the Paths That End in “Cold Enough”

The beer can reach the target temperature by any of the three methods. These paths are disjoint because each trial uses only one method.

$$P(R \cap C) = P(R)P(C | R) = (0.30)(0.10) = 0.030$$

$$P(F \cap C) = P(F)P(C | F) = (0.30)(0.55) = 0.165$$

$$P(I \cap C) = P(I)P(C | I) = (0.40)(0.85) = 0.340$$

$$P(R \cap C^c) = P(R)P(C^c | R) = (0.30)(0.90) = 0.27$$

$$P(F \cap C^c) = P(F)P(C^c | F) = (0.30)(0.45) = 0.135$$

$$P(I \cap C^c) = P(I)P(C^c | I) = (0.40)(0.15) = 0.06$$

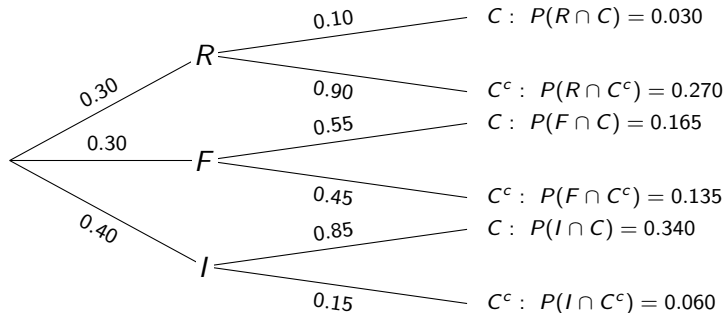
So

$$P(C) = 0.030 + 0.165 + 0.340 = \boxed{0.535}$$

$$P(C) = 0.27 + 0.135 + 0.06 = \boxed{0.465}$$

Under this model, the beer is cold enough before the pizza arrives in 53.5% of trials.

Built Tree: Is the Beer Cold Enough?



Now Reverse the Condition

We were given

$$P(C | I) = 0.85$$

This says that **among trials using an ice-water bath** 85% reach the target temperature.

A different question

Suppose the beer **did** reach the target temperature. What is the probability that it was cooled in an ice-water bath?

Now we want $P(I | C)$. The group we must consider is **all trials that ended with cold enough beer**, regardless of the method used.

$$P(C | I) \quad \text{versus} \quad P(I | C)$$

Calculate the Reverse Conditional Probability

The tree gives us

$$P(I \cap C) = 0.340 \quad \text{and} \quad P(C) = 0.535$$

Among trials where the beer was cold enough, the fraction that used an ice-water bath is

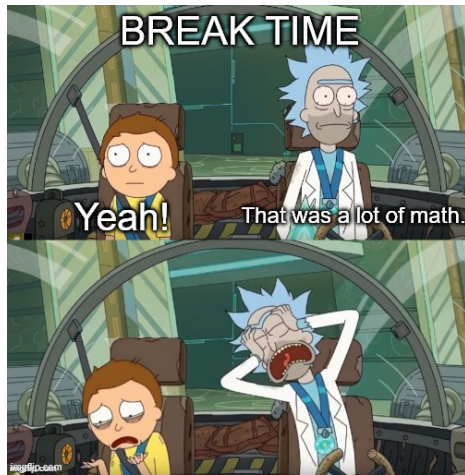
$$P(I \mid C) = \frac{P(I \cap C)}{P(C)} = \frac{0.340}{0.535} \approx \boxed{0.636}$$

Interpretation

Of the trials that ended with cold enough beer, about 63.6% used an ice-water bath.

Knowing that an ice-water bath was used gives $P(C \mid I) = 0.85$. Knowing that the beer became cold gives $P(I \mid C) \approx 0.636$. The next section develops a general way to reverse such probabilities.

Break Time



Why Bayes's Theorem Is Needed

Often we know a probability in one direction but need the reverse.

$$P(\text{Positive} \mid \text{Disease}) \quad \text{versus} \quad P(\text{Disease} \mid \text{Positive})$$

These are not the same.

Diagnostic-Test Language

- **Sensitivity:** $P(\text{Test}^+ \mid \text{Disease})$
- **Specificity:** $P(\text{Test}^- \mid \text{No Disease})$
- **Positive Predictive Value:** $P(\text{Disease} \mid \text{Test}^+)$

Example 10.14: Mammography

Let C mean that a patient has breast cancer, and let $+$ and $-$ denote positive and negative screening results. For a patient in their 60s, suppose our model gives

$$P(C) = 0.0365 \quad P(C^c) = 1 - 0.0365 = 0.9635$$

The screening test has

$$P(+ | C) = 0.85 \quad \text{and} \quad P(- | C^c) = 0.95$$

Thus, its false-positive probability is

$$P(+ | C^c) = 1 - 0.95 = 0.05$$

Question

If the patient receives a **positive result**, what is the probability that they have breast cancer, $P(C | +)$?

Example 10.14: Mammography

The test information includes its **sensitivity**: $P(+ | C) = 0.85$

This is the probability of a positive result *among patients who have cancer*. The question asks for

$$P(C | +)$$

the probability of cancer *among patients who receive a positive result*.

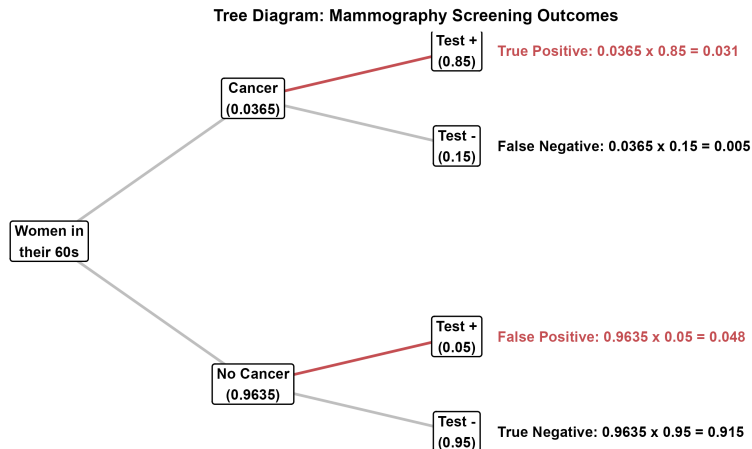
Where can a positive result come from?

A positive result can occur in either of two disjoint groups:

$$C \cap + \quad \text{or} \quad C^c \cap +$$

We need both groups to find the probability of a positive result.

Example 10.14: Mammography



Follow the branches from cancer status to test result. Multiplying along a path gives its joint probability.

Example 10.14: Mammography

Cancer and a positive result:

$$P(C \cap +) = P(C)P(+ \mid C) = (0.0365)(0.85) = 0.031025$$

No cancer and a positive result:

$$P(C^c \cap +) = P(C^c)P(+ \mid C^c) = (0.9635)(0.05) = 0.048175$$

All positive results

The paths are disjoint so

$$P(+) = 0.031025 + 0.048175 = 0.0792$$

Example 10.14: Mammography

We now restrict attention to patients with **positive results**. Of all positive results, the cancer-and-positive path accounts for 0.031025:

$$P(C \mid +) = \frac{P(C \cap +)}{P(+)} = \frac{0.031025}{0.0792} \approx \boxed{0.3917}$$

Interpretation

Under this model, about 39.2% of patients who receive a positive screening result have breast cancer. This probability is called the **positive predictive value**.

The 85% sensitivity and the 39.2% positive predictive value condition on different groups.

The Law of Total Probability

Suppose $A_1, \dots, A_k$ form a partition of the sample space: they are disjoint and exhaustive.

Total Probability

$$P(B) = \sum_{j=1}^k P(B \mid A_j)P(A_j)$$

For mammography

$$P(+) = P(+ \mid C)P(C) + P(+ \mid C^c)P(C^c)$$

This denominator represents **all possible paths that lead to a positive test**.

The Formal Bayes's Theorem

If $A_1, \dots, A_k$ form a partition and $P(B) > 0$, then

Bayes's Theorem

$$P(A_i | B) = \frac{P(B | A_i)P(A_i)}{\sum_{j=1}^k P(B | A_j)P(A_j)}$$

- **Prior:** $P(A_i)$ is our probability for A_i before observing B .
- **Likelihood:** $P(B | A_i)$ tells us how likely the observation B is if A_i is true.
- **Posterior:** $P(A_i | B)$ is the updated probability for A_i after observing B .
- **Evidence:** $\sum_{j=1}^k P(B | A_j)P(A_j) = P(B)$ accounts for every way B can occur.

Connection to the Tree Diagram

The tree supplies the prior and likelihood for each path. Bayes's Theorem uses them to update the probability of A_i .

Example 10.14: Mammography

Question

Given a positive mammogram, what is the probability of breast cancer, $P(C \mid +)$?

For two underlying groups, cancer and no cancer,

$$P(C \mid +) = \frac{P(+ \mid C)P(C)}{P(+ \mid C)P(C) + P(+ \mid C^c)P(C^c)}$$

- The **numerator** is the probability of the cancer-and-positive path.
- The **denominator** is the probability of **all ways to obtain a positive test**.

Same Logic as the Tree

Bayes's formula is not a new kind of reasoning. It packages the tree-diagram calculation into one equation.

Example 10.14: Mammography

Substitute the model probabilities:

$$P(C \mid +) = \frac{(0.85)(0.0365)}{(0.85)(0.0365) + (0.05)(0.9635)} = \frac{0.031025}{0.031025 + 0.048175} = \frac{0.03102}{0.0792} = \mathbf{0.3917}$$

Keep the Question in View

We are asking: among all positive tests, what fraction came from the cancer branch?

The Base-Rate Effect

Bayes's Theorem combines two kinds of information:

- the **base rate or prior probability**; and
- how strongly the observed evidence is associated with each underlying state.

When the condition is rare, even a fairly good test can have a low positive predictive value because the much larger negative population can produce many false positives.

Next Examples

Lyme disease, environmental lead screening, and polygraph classification all use the same mathematical structure.

Bayes Example: Lyme Disease

Let L mean that a person has Lyme disease, and let $+$ mean a positive test result. Suppose a hypothetical region and test have the following probabilities:

$$P(L) = 0.02 \quad P(+ \mid L) = 0.90 \quad P(+ \mid L^c) = 0.20$$

The probability of not having Lyme disease is

$$P(L^c) = 1 - 0.02 = 0.98$$

Question

If a person receives a **positive result**, what is the probability that they have Lyme disease, $P(L \mid +)$?

Bayes Example: Lyme Disease

A positive result can come from a person with Lyme disease or a person without it. Bayes's Theorem includes both paths:

$$P(L \mid +) = \frac{P(L)P(+ \mid L)}{P(L)P(+ \mid L) + P(L^c)P(+ \mid L^c)}$$

Substituting the model probabilities gives

$$\begin{aligned} P(L \mid +) &= \frac{(0.02)(0.90)}{(0.02)(0.90) + (0.98)(0.20)} \\ &= \frac{0.018}{0.018 + 0.196} \\ &\approx \boxed{0.084} \end{aligned}$$

Interpretation

Under this hypothetical model, about 8.4% of people with a positive result have Lyme disease.

Bayes Example: Lead Screening

Let L mean that a water sample contains dangerous lead levels and let R mean that the screening reagent turns red. Suppose the following hypothetical probabilities apply:

$$P(L) = 0.05 \quad P(R | L) = 0.95 \quad P(R | L^c) = 0.10$$

The probability that a sample does not contain dangerous lead levels is

$$P(L^c) = 1 - 0.05 = 0.95$$

Question

If a sample **turns red**, what is the probability that it contains dangerous lead levels, $P(L | R)$?

Bayes Example: Lead Screening

A red result can come from a sample with dangerous lead levels or one without them. Bayes's Theorem includes both paths:

$$P(L | R) = \frac{P(L)P(R | L)}{P(L)P(R | L) + P(L^c)P(R | L^c)}$$

Substituting the model probabilities gives

$$\begin{aligned} P(L | R) &= \frac{(0.05)(0.95)}{(0.05)(0.95) + (0.95)(0.10)} \\ &= \frac{0.0475}{0.0475 + 0.0950} = \boxed{\frac{1}{3} \approx 0.333} \end{aligned}$$

Interpretation

Under this hypothetical model, about one third of samples that turn red contain dangerous lead levels.

Bayes Example: Polygraph Classification

Let L mean that a person is lying about the matter being tested, and let $+$ mean a positive deception indication. For a hypothetical group of people being tested, suppose

$$P(L) = 0.10 \quad P(+ | L) = 0.80 \quad P(+ | L^c) = 0.15$$

The probability that a person is telling the truth is

$$P(L^c) = 1 - 0.10 = 0.90$$

Question

If a person receives a **positive deception indication**, what is the probability that they are lying, $P(L | +)$?

Bayes Example: Polygraph Classification — Solution

A positive indication can come from a person who is lying or a person who is telling the truth. Bayes's Theorem includes both paths:

$$P(L \mid +) = \frac{P(L)P(+ \mid L)}{P(L)P(+ \mid L) + P(L^c)P(+ \mid L^c)}$$

Substituting the hypothetical probabilities gives

$$\begin{aligned} P(L \mid +) &= \frac{(0.10)(0.80)}{(0.10)(0.80) + (0.90)(0.15)} \\ &= \frac{0.080}{0.080 + 0.135} \approx \boxed{0.372} \end{aligned}$$

Interpretation

Under this hypothetical model, about 37.2% of people who receive a positive indication are lying about the matter tested.

Playing Blackjack

Question

After ten known non-10-valued cards are removed from a standard deck, what is the probability that the next card has value 10?

In a standard 52-card deck, 16 cards have value 10 (10, J, Q, K), so initially

$$P(10) = \frac{16}{52} = \mathbf{0.307}$$

Suppose ten known non-10-valued cards are removed.

- The deck now contains only 42 cards.
- All 16 ten-valued cards are still present.

Why This Is Dependence

The first ten observed cards changed the composition of the remaining deck, so the probability of the next card depends on what was previously removed.

Playing Blackjack

After the ten non-10-valued cards have been removed

$$P(10 \mid 10 \text{ non-10 cards removed}) = \frac{16}{42} = \mathbf{0.381}$$

The probability increased from 30.7% to 38.1% because the denominator decreased while all 16 favourable ten-valued cards remained.

Learning Purpose

Sampling without replacement creates dependence. Conditional probability updates the chance of future draws as the composition of the remaining population changes.

This change in one card probability alone does not establish the expected advantage of a complete blackjack strategy.

Chapter 10 Synthesis: Which Probability Tool?

Question	Key Idea	Useful Rule
Can A and B occur together?	Disjointness	Check $P(A \cap B) = 0$
Chance of A or B ?	Union	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Does knowing A change B ?	Independence	Compare $P(B A)$ with $P(B)$
Chance of B given A ?	Conditioning	$P(B A) = P(A \cap B) / P(A)$
Chance of both?	General multiplication	$P(A \cap B) = P(A)P(B A)$
Several stages?	Tree diagram	Multiply paths; add disjoint paths
Reverse a conditional?	Bayes	Use priors and all paths to the evidence